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Deep Generative Models

Lecture 14

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Recap: Discrete Diffusion (Lecture 12)

ELBO

$$\begin{aligned}\mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

Recap: Discrete Diffusion (Lecture 12)

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$

- Both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $q(\mathbf{x}_t|\mathbf{x}_0)$ are known analytically from the forward process.
- The reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a learned categorical distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}(\boldsymbol{\pi}_\theta(\mathbf{x}_t, t)).$$

Recap: Discrete Diffusion (Lecture 12)

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}} \left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \right).$$

Categorical KL

$$D_{\text{KL}}(\text{Cat}(\mathbf{q}) \parallel \text{Cat}(\mathbf{p})) = \sum_{k=1}^K q_k \log \frac{q_k}{p_k} = H(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}),$$

- $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$ is a constant w.r.t. θ .
- $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$ is a **cross-entropy loss**.

Therefore, minimizing $\mathcal{L}_t$ w.r.t. θ is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} H \left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0), p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \right).$$

Setup: where PF-ODE solvers floor

Recall: probability flow ODE

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \mathbf{s}(\mathbf{x}, t) \right] dt, \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x}).$$

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- Plugging s_θ into the PF-ODE and using a high-order ODE solver (Heun, RK) brings the number of steps from 100–1000 down to 20–50.

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- Plugging s_θ into the PF-ODE and using a high-order ODE solver (Heun, RK) brings the number of steps from 100–1000 down to 20–50.
- Each step still costs one network call — truncation error compounds along the trajectory.
- Can we go further — to **1–4** steps?

Setup: shortcut networks

- Idea: **learn a shortcut network** that compresses many ODE-solver steps into one.
- It maps any $(\mathbf{x}_t, t)$ directly to the data-side endpoint of the same PF-ODE trajectory — one network call instead of many.

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Two approaches in this lecture

- **Consistency Models** — enforce *self-consistency* along PF-ODE trajectories (per-sample mapping).
- **Distribution Matching Distillation** — match the *distributions* of generator and data via score differences (population-level).

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Both reuse a pretrained score s_ϕ as a teacher.

Outline

01 Consistency Models

02 Distribution Matching Distillation

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02 Distribution Matching Distillation

Consistency function: an ODE perspective

Goal

Learn a one-step generator $f_\theta : (\mathbf{x}_t, t) \mapsto \mathbf{x}_\epsilon$ that pulls any noisy $\mathbf{x}_t$ on a PF-ODE trajectory back to its data endpoint $\mathbf{x}_\epsilon$, $t \in [\epsilon, T]$.

$\epsilon > 0$ is a small numerical cutoff — the score $s(\mathbf{x}, t)$ blows up as $t \rightarrow 0$, so we stop one ϵ -step short and treat $\mathbf{x}_\epsilon \approx \mathbf{x}_0$ as the data sample.

Consistency function: an ODE perspective

Specification: ODE problem along the trajectory

$$\underbrace{\frac{d}{dt} f_{\theta}(\mathbf{x}_t, t) = 0}_{\text{differential form: constant along PF-ODE}}, \quad \underbrace{f_{\theta}(\mathbf{x}_{\epsilon}, \epsilon) = \mathbf{x}_{\epsilon}}_{\text{boundary: value pinned at } t=\epsilon}.$$

Consistency function: an ODE perspective

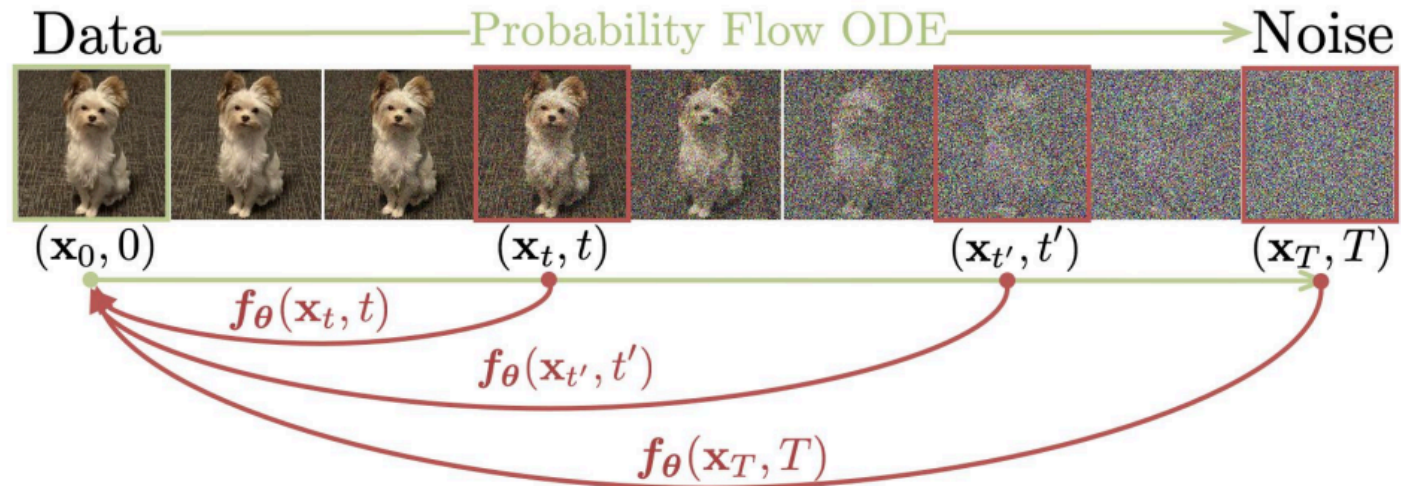
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Together the two conditions pin f_{θ} uniquely: a function that is constant on each trajectory and equals $\mathbf{x}_{\epsilon}$ at the data-side end must equal $\mathbf{x}_{\epsilon}$ *everywhere* on that trajectory.

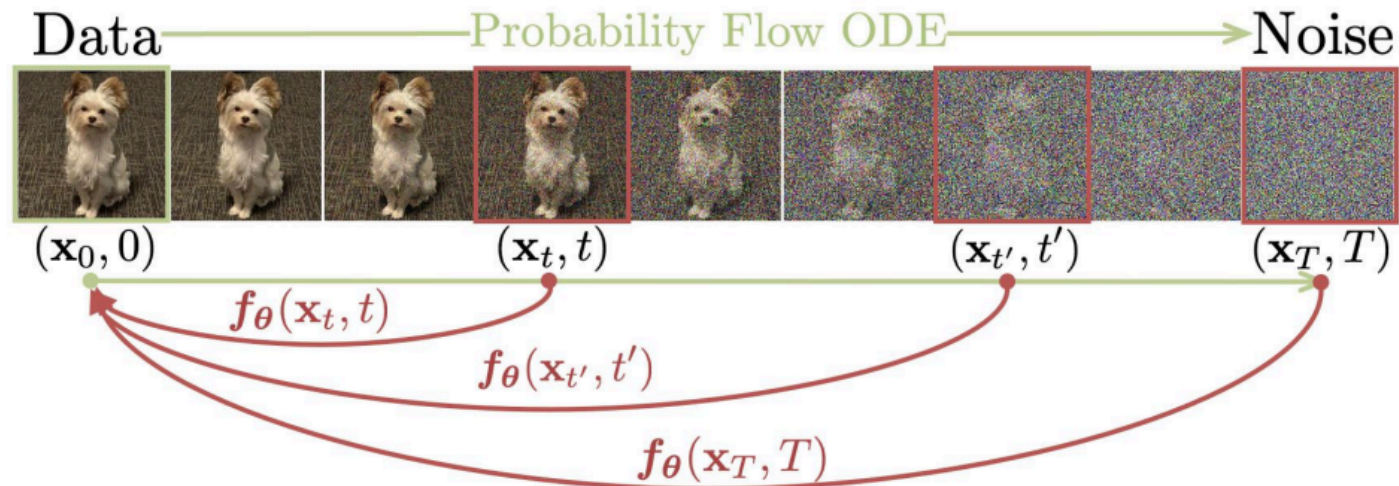
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Consistency function: an ODE perspective

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- To optimize the differential condition $\frac{d}{dt} f_{\theta}(\mathbf{x}_t, t) = 0$ we need a derivative through the network.
- Integrate it between two times t, t' on the *same* PF-ODE trajectory.

Self-consistency: the integrated form

Idea is to replace $df_{\theta}/dt = 0$ by a finite difference between two adjacent points.

Self-consistency property

$$f_{\theta}(\mathbf{x}_t, t) = f_{\theta}(\mathbf{x}_{t'}, t'), \quad \forall t, t' \in [\epsilon, T],$$

whenever $\mathbf{x}_t$ and $\mathbf{x}_{t'}$ lie **on the same PF-ODE trajectory**.

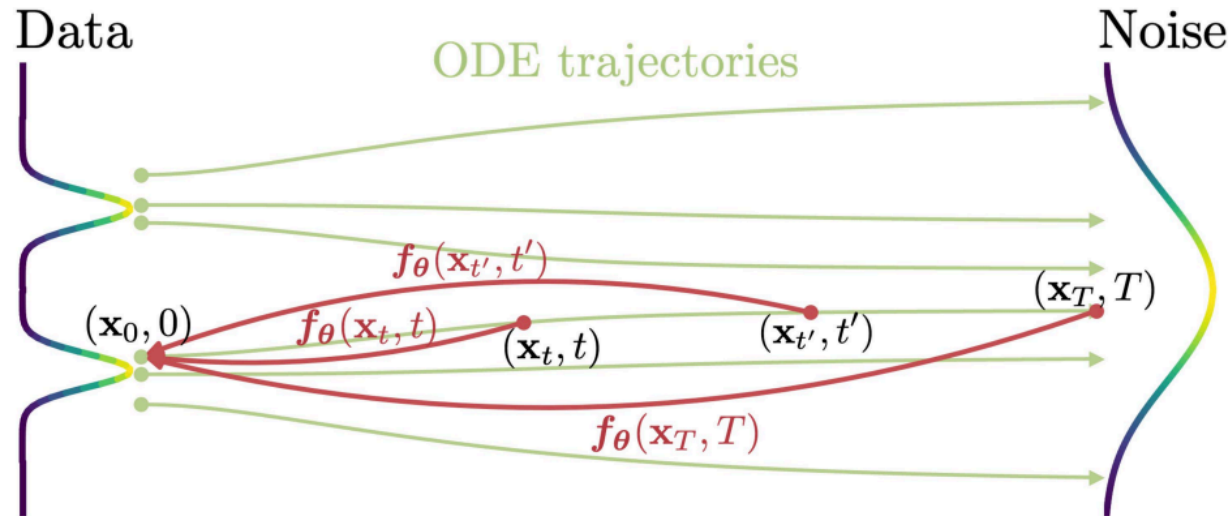
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Skip-connection parameterization

Parameterization

We parametrize the consistency model to guarantee the boundary condition $f_{\theta}(\mathbf{x}_{\epsilon}, \epsilon) = \mathbf{x}_{\epsilon}$

$$f_{\theta}(\mathbf{x}, t) = c_{\text{skip}}(t) \mathbf{x} + c_{\text{out}}(t) F_{\theta}(\mathbf{x}, t).$$

- $F_{\theta}(\mathbf{x}, t)$ — free-form neural network.
- $c_{\text{skip}}(t)$, $c_{\text{out}}(t)$ — differentiable scalar weights with $c_{\text{skip}}(\epsilon) = 1$ and $c_{\text{out}}(\epsilon) = 0$.

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At $t = \epsilon$ the skip term passes $\mathbf{x}$ through and the network term is killed: $f_{\theta}(\mathbf{x}_{\epsilon}, \epsilon) = \mathbf{x}_{\epsilon}$ holds *structurally*, for any θ .

Consistency Distillation (CD)

Enforce $df_{\theta}/dt = 0$ as a *finite difference* on adjacent points of the teacher's PF-ODE trajectory. Discretize $[\epsilon, T]$ as $\epsilon = t_1 < \dots < t_N = T$; $s_{\phi}(\mathbf{x}, t)$ — pretrained teacher score.

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One ODE-solver step on the empirical PF-ODE

$$\hat{\mathbf{x}}_{t_n}^\phi = \mathbf{x}_{t_{n+1}} + (t_n - t_{n+1}) \Phi(\mathbf{x}_{t_{n+1}}, t_{n+1}; \phi),$$

one step of `ODESolve` for $d\mathbf{x}/dt = -t \cdot s_\phi(\mathbf{x}, t)$ (Euler: $\Phi = -t s_\phi$; Heun: average of two stages). The RHS follows from the VE-SDE PF-ODE with $\sigma(t) = t$ ($\mathbf{x}_t = \mathbf{x}_0 + t \epsilon$): then $\mathbf{f} \equiv 0$, $g^2(t) = 2t$, so $\frac{1}{2}g^2(t) = t$.

Consistency Distillation (CD)

CD loss

$$\mathcal{L}_{\text{CD}}(\boldsymbol{\theta}) = \mathbb{E} \left[\lambda(t_n) d(f_{\boldsymbol{\theta}}(\mathbf{x}_{t_{n+1}}, t_{n+1}), f_{\boldsymbol{\theta}^-}(\hat{\mathbf{x}}_{t_n}^{\phi}, t_n)) \right].$$

$\lambda(t_n) > 0$ — weighting; $d(\mathbf{u}, \mathbf{v}) \in \{\|\mathbf{u} - \mathbf{v}\|_2^2, \|\mathbf{u} - \mathbf{v}\|_1, \text{LPIPS}\}$; $\boldsymbol{\theta}^-$ — EMA of $\boldsymbol{\theta}$ (stop-gradient target, prevents collapse).

Pseudocode (Consistency Distillation)

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $n \sim U\{1, N-1\}$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute $\mathbf{x}_{t_{n+1}} = \mathbf{x}_0 + t_{n+1} \epsilon$.
3. Solve one teacher ODE step $\hat{\mathbf{x}}_{t_n}^\phi = \mathbf{x}_{t_{n+1}} + (t_n - t_{n+1}) \Phi(\mathbf{x}_{t_{n+1}}, t_{n+1}; \phi)$.
4. Compute loss $\mathcal{L} = \lambda(t_n) d(f_\theta(\mathbf{x}_{t_{n+1}}, t_{n+1}), f_{\theta^-}(\hat{\mathbf{x}}_{t_n}^\phi, t_n))$.
5. Update EMA target $\theta^- \leftarrow \mu \theta^- + (1 - \mu) \theta$.

Pseudocode (Consistency Distillation)

Sampling (Multi-step)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, T^2 \mathbf{I})$; denoise $\mathbf{x}_0 \leftarrow f_{\theta}(\mathbf{x}_T, T)$.
2. For $i = 1, \dots, M-1$: sample $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, compute $\mathbf{x}_{\tau_i} = \mathbf{x}_0 + \tau_i \epsilon$, denoise $\mathbf{x}_0 \leftarrow f_{\theta}(\mathbf{x}_{\tau_i}, \tau_i)$.
3. Return $\mathbf{x}_0$.

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Multi-step trades a few extra NFEs ($M = 2-4$) for higher quality; $M = 1$ recovers single-step sampling.

Latent Consistency Models (LCM)

Apply Consistency Distillation to a latent diffusion model (Stable Diffusion).

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Guided consistency distillation

Absorb classifier-free guidance into f_θ : an augmented function $f_\theta(\mathbf{x}_t, t, \omega, \mathbf{c})$ takes the guidance scale ω as input. The teacher uses CFG:

$$\tilde{\mathbf{s}}_\phi(\mathbf{x}_t, t, \omega, \mathbf{c}) = (1 + \omega) \mathbf{s}_\phi(\mathbf{x}_t, t, \mathbf{c}) - \omega \mathbf{s}_\phi(\mathbf{x}_t, t, \emptyset).$$

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Skipping-step

Enforce consistency over k steps instead of 1:

$$f_\theta(\mathbf{x}_{t_{n+k}}, t_{n+k}) \approx f_{\theta^-}(\hat{\mathbf{x}}_{t_n}, t_n).$$

Larger SNR gap $\Rightarrow$ stronger training signal.

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One-step generator

Setup

Train a generator $G_\theta : \mathbf{z} \mapsto \mathbf{x}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

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Distribution matching, not per-sample matching.

- CM ties each $\mathbf{z}$ to a specific PF-ODE endpoint.
- DMD lets G_θ re-organize the noise-to-image map — $\mathbf{z}$ and $G_\theta(\mathbf{z})$ need not lie on a teacher trajectory.

KL gradient via score difference

Diffuse the generator output: $\mathbf{x}_t = G_{\theta}(\mathbf{z}) + \sigma_t \epsilon$. Let $p_{\text{fake},t}$ and $p_{\text{real},t}$ be the corresponding marginals.

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Variational score-distillation identity

$$\begin{aligned} \nabla_{\theta} D_{\text{KL}}(p_{\text{fake},t} \parallel p_{\text{real},t}) \\ = - \mathbb{E} \left[(\mathbf{s}_{\text{real},t}(\mathbf{x}_t) - \mathbf{s}_{\text{fake},t}(\mathbf{x}_t)) \cdot \nabla_{\theta} G_{\theta}(\mathbf{z}) \right]. \end{aligned}$$

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- Likelihood-free: only score *differences* matter.
- Integrate over t with weighting $w(t) \Rightarrow$ tractable training signal.

Two score networks

Real score (frozen)

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Alternating optimization

1. Update s_{ψ} on a batch from G_{θ} (score matching).
2. Update G_{θ} with the score-difference gradient.

DMD1: total loss

Total objective

$$\mathcal{L}_{\text{DMD}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{KL-grad}}(\boldsymbol{\theta}) + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}(\boldsymbol{\theta}).$$

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Regression term

$$\mathcal{L}_{\text{reg}}(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{z}, \mathbf{x}^*)} \|G_{\boldsymbol{\theta}}(\mathbf{z}) - \mathbf{x}^*\|^2,$$

on a few hundred precomputed $(\mathbf{z}, \mathbf{x}^*)$ pairs ($\mathbf{x}^*$ = teacher's PF-ODE output for $\mathbf{z}$).

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Interpretation: regression anchors $G_{\boldsymbol{\theta}}$ near the teacher's deterministic map; the KL-gradient term then reshapes the *distribution* of outputs.

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- **Multi-step generator**: G_θ becomes a 4-step network, recovering quality lost at the strict 1-step setting.

Closes most of the quality gap between distilled and full-step diffusion.

Summary

- PF-ODE solvers (Heun, RK) floor at ~ 20 steps; further acceleration requires **learning a shortcut network**.
- **Consistency Models**: a neural network f_θ specified by $df_\theta/dt = 0$ along PF-ODE + boundary $f_\theta(\mathbf{x}_\epsilon, \epsilon) = \mathbf{x}_\epsilon$; CD enforces the differential condition as a finite-difference loss on a teacher's trajectory.
- LCM extends CD to latent SD with guided distillation and skipping-step.
- **DMD**: match generator and data *distributions* via the score-difference gradient; two score networks (frozen real, online fake) in alternating optimization. DMD2 adds GAN, drops regression, adds multi-step.
- Tradeoffs: CM is more principled, DMD currently delivers the best 1-step quality.